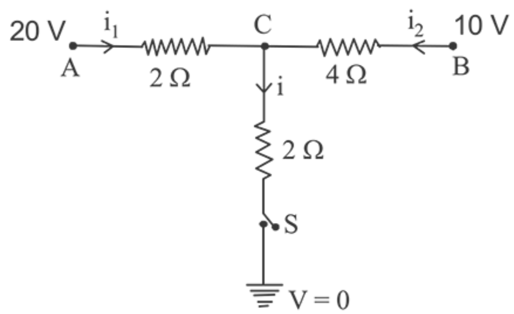
# JEE Mains 2019 Chapter wise Question Bank

## Current Electricity - Questions

**Q1**

When the switch S, in the circuit shown, is closed then the valued of current $i$ will be:

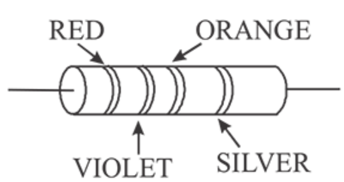



(1) 3A (2) 5A
(3) 4A (4) 2A

**9 Jan Morning**

**Q2**

A resistance is shown in the figure. Its value and tolerance are given respectively by:




(1) 270 Ω, 10% (2) 27 kΩ, 10%
(3) 27 kΩ, 20% (4) 270 Ω, 5%

**9 Jan Morning**

**Q3**

Drift speed of electrons, when 1.5 A of current flows in a copper wire of cross section 5 $mm^2$, is $v$. If the electron density in copper is $9 \times 10^{28}/m^3$ the value of $v$ in mm/s close to (Take charge of electron to be = $1.6 \times 10^{-19}$C)

(1) 0.02 (2) 3
(3) 2 (4) 0.2

**9 Jan Morning**

**Q4**

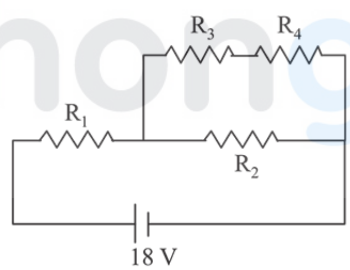
A carbon resistance has following colour code. What is the value of the resistance?


(1) 530 kΩ ± 5% (2) 5.3 kΩ ± 5%
(3) 6.4 MΩ ± 5% (4) 64 MΩ ± 10%

**9 Jan Evening**

**Q5**

In the given circuit the internal resistance of the 18 V cell is negligible. If $R_1$ = 400Ω, $R_3$ = 100 Ω and $R_4$ = 500 Ω and the reading of an ideal voltmeter across $R_4$ is 5V, then the value of $R_2$ will be:




(1) 300 Ω (2) 450 Ω
(3) 550 Ω (4) 230 Ω

**9 Jan Evening**

**Q6**

A uniform metallic wire has a resistance of 18 Ω and is bent into an equilateral triangle. Then, the resistance between any two vertices of the triangle is:
(1) 4 Ω (2) 8 Ω (3) 12 W (4) 2 W

**10 Jan Morning**

**Q7**

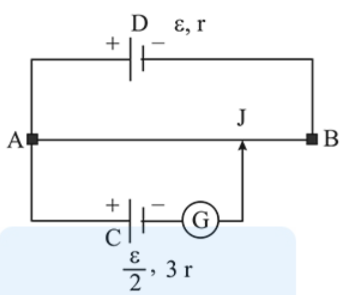
A 2 W carbon resistor is color coded with green, black, red and brown respectively. The maximum current which can be passed through this resistor is:

(1) 20 mA (2) 100 mA
(3) 0.4 mA (4) 63 mA

**10 Jan Morning**

### Q8

A potentiometer wire AB having length L and resistance 12 r is joined to a cell D of emf ε and internal resistance r. A cell C having emf ε/2 and internal resistance 3r is connected. The length AJ at which the galvanometer as shown in fig. shows no deflection is:

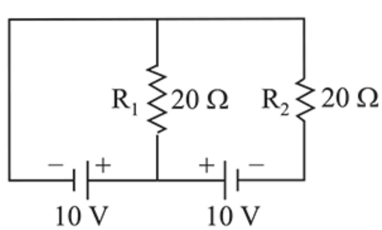



(1) $\frac{11}{12}L$ (2) $\frac{11}{24}L$ (3) $\frac{13}{24}L$ (4) $\frac{5}{12}L$

**10 Jan Morning**

### Q9

In the given circuit the cells have zero internal resistance. The currents (in Amperes) passing through resistance $R_1$ and $R_2$ respectively, are:

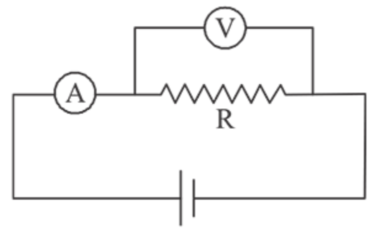



(1) 1, 2 (2) 2, 2
(3) 0.5, 0 (4) 0, 1

**10 Jan Morning**

### Q10

The actual value of resistance R, shown in the figure is 30 Ω. This is measured in an experiment as shown using the standard formula $R = \frac{V}{I}$, where V and I are the reading of the voltmeter and ammeter, respectively. If the measured value of R is 5% less, then the internal resistance of the voltmeter is:




(1) 600 Ω (2) 570 Ω
(3) 35 Ω (4) 350 Ω

**10 Jan Evening**

### Q11

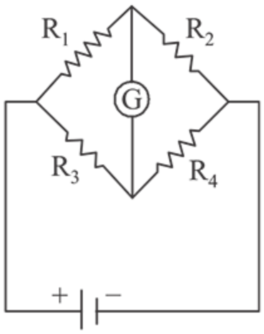
A current of 2 mA was passed through an unknown resistor which dissipated a power of 4.4 W. Dissipated power when an ideal power supply of 11 V is connected across it is:

(1) $11 \times 10^{-5}$ W (2) $11 \times 10^{-3}$ W
(3) $11 \times 10^{-4}$ W (4) $11 \times 10^{5}$ W

**10 Jan Evening**

### Q11

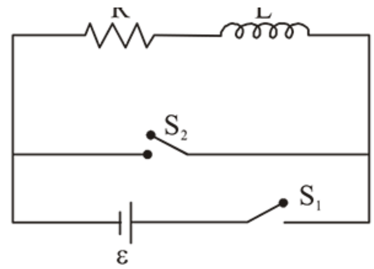
The Wheatstone bridge shown in Fig. here, gets balanced when the carbon resistor used as $R_1$ has the colour code (Orange, Red, Brown). The resistors $R_2$ and $R_4$ are 80 Ω and 40 Ω, respectively.
Assuming that the colour code for the carbon resistors gives their accurate values, the colour code for the carbon resistor, used as $R_3$, would be:



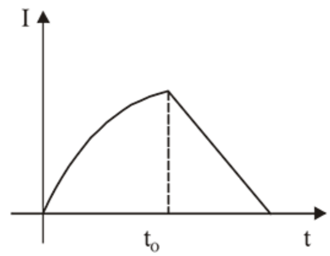

(1) Brown, Blue, Brown (2) Brown, Blue, Black
(3) Red, Green, Brown (4) Grey, Black, Brown

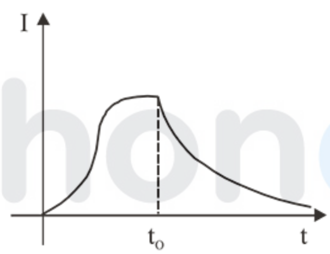
**10 Jan Evening**

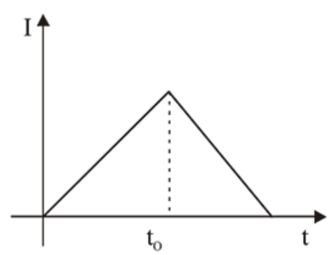
**Q12**

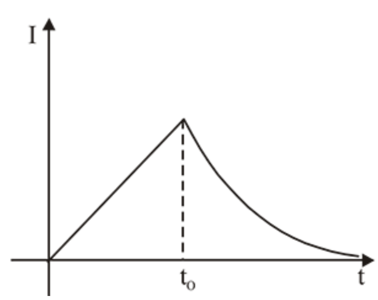



the switch $S_1$ is closed at time $t = 0$ and the switch $S_2$ is kept open. At some later time ($t_0$), the switch $S_1$ is opened and $S_2$ is closed. the behaviour of the current I as a function of time 't' is given by:

(1)



(2)


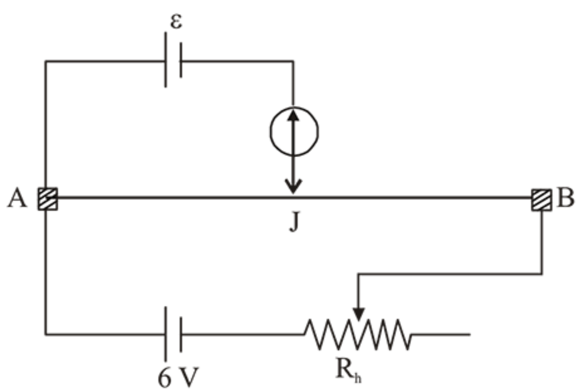

(3)



(4)



**11 Jan Morning**

**Q13**

The resistance of the meter bridge AB in given figure is $4\Omega$. With a cell of emf $\varepsilon = 0.5\text{ V}$ and rheostat resistance $R_h = 2\Omega$ the null point is obtained at some point J. When the cell is replaced by another one of emf $\varepsilon = \varepsilon_2$ the same null point J is found for $R_h = 6\Omega$. The emf $\varepsilon_2$ is,:




(1) 0.4 V (2) 0.3 V
(3) 0.6 V (4) 0.5 V

**11 Jan Morning**

**Q14**

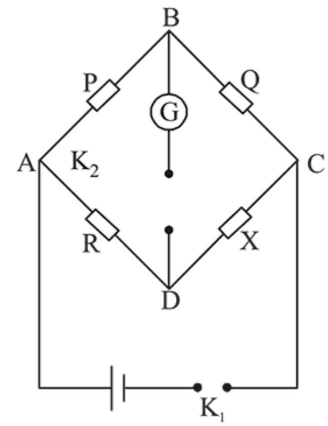
Two equal resistances when connected in series to a battery, consume electric power of 60 W. If these resistance are now connected in parallel combination to the same battery, the electric power consumed will be :

(1) 60 W (2) 240 W
(3) 120 W (4) 30 W

**11 Jan Morning**

**Q14**

In a Wheatstone bridge (see fig.), Resistances P and Q are approximately equal. When $R = 400\,\Omega$, the bridge is balanced. On interchanging P and Q, the value of R, for balance, is $405\Omega$. The value of X is close to :

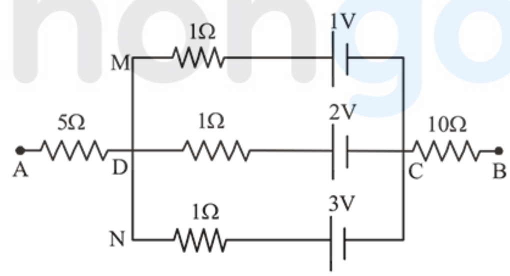



(1) 401.5 ohm (2) 404.5 ohm
(3) 403.5 ohm (4) 402.5 ohm

**11 Jan Morning**

**Q15**

In the circuit shown, the potential difference between A and B is :




(1) 1 V (2) 2 V
(3) 3 V (4) 6 V

**11 Jan Evening**

**Q16**

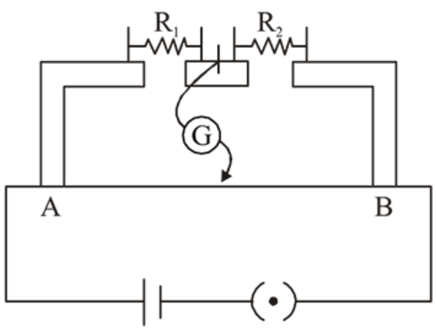
A galvanometer having a resistance of $20\,\Omega$ and 30 division on both sides has figure of merit 0.005 ampere/division. The resistance that should be connected in series such that it can be used as a voltmeter upto 15 volt, is:

(1) $100\,\Omega$ (2) $120\,\Omega$
(3) $80\,\Omega$ (4) $125\,\Omega$

**11 Jan Evening**

**Q17**

In the experimental set up of metre bridge shown in the figure, the null point is obtaine data distance of 40 cm from A. If a 10 Ω resistor is connected in series with $R_1$, the null point shifts by 10 cm. The resistance that should be connected in parallel with $(R_1 + 10)$ Ω such that the null point shifts back to its initial position is :




(1) 20 Ω (2) 40 Ω
(3) 60 Ω (4) 30 Ω

**11 Jan Evening**

**Q18**

Two electric bulbs, rated at (25 W, 220 V) and (100 W, 220 V), are connected in series across a 220 V voltage source. If the 25 W and 100 W bulbs draw powers $P_1$ and $P_2$ respectively, then:

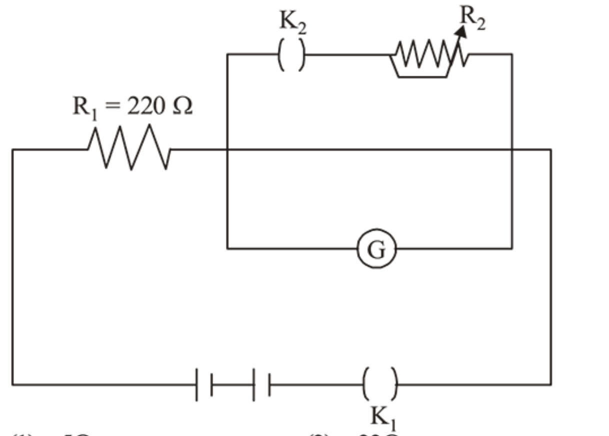
(1) $P_1$=16 W, $P_2$=4 W (2) $P_1$=16 W, $P_2$=9 W
(3) $P_1$=9 W, $P_2$=16 W (4) $P_1$=4 W, $P_2$=16 W

1

**12 Jan Morning**

**Q19**

The galvanometer deflection, when key $K_1$ is closed but $K_2$ is open, equals $\theta_0$ (see figure). On closing $K_2$ also and adjusting $R_2$ to 5Ω, the deflection in galvanometer becomes $\frac{\theta_0}{5}$. The resistance of the galvanometer is, then, given by [Neglect the internal resistance of battery]:

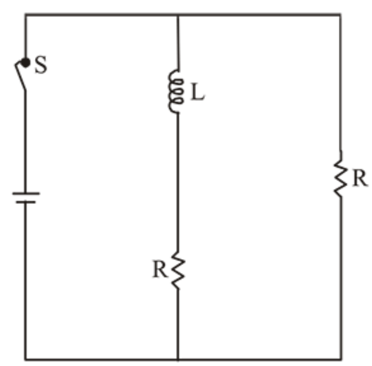



(1) 5Ω (2) 22Ω
(3) 25Ω (4) 12Ω

**12 Jan Morning**

**Q20**

In the figure shown, a circuit contains two identical resistors with resistance R = 5 Ω and an inductance with L = 2 mH. An ideal battery of 15 V is connected in the circuit. What will be the current through the battery long after the switch is closed?

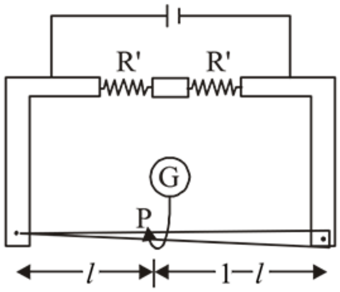



(1) 5.5 A (2) 7.5 A
(3) 3 A (4) 6 A

**12 Jan Morning**

**Q21**

In a meter bridge, the wire of length 1 m has a non-uniform cross-section such that, the variation $\frac{dR}{dl}$ of its resistance R with length $l$ is $\frac{dR}{dl} \propto \frac{1}{\sqrt{l}}$. Two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P. What is the length AP?




(1) 0.2m (2) 0.3m
(3) 0.25m (4) 0.35m

**12 Jan Morning**

**Q22**

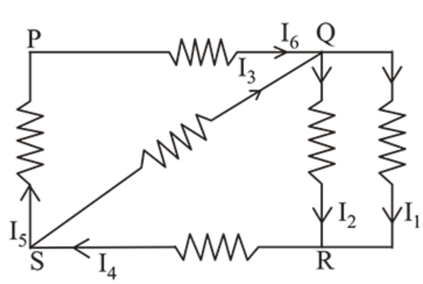
An ideal battery of 4 V and resistance R are connected in series in the primary circuit of a potentionmeter of length 1 m and resistance 5Ω. The value of R, to give a potential difference of 5 mV across 10 cm of potentiometer wire is:

(1) 490Ω (2) 480Ω
(3) 395Ω (4) 495Ω

**12 Jan Morning**

**Q23**

In the given circuit diagram, the currents, $I_1 = -0.3$ A, $I_4 = 0.8$ A and $I_5 = 0.4$ A, are flowing as shown. The currents $I_2$, $I_3$ and $I_6$, respectively, are :




(1) 1.1 A, –0.4 A, 0.4 A (2) 1.1 A, 0.4 A, 0.4 A
(3) 0.4 A, 1.1 A, 0.4 A (4) –0.4 A, 0.4 A, 1.1 A

**12 Jan Evening**

**Q24**

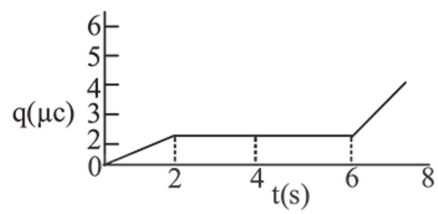
A galvanometer, whose resistance is 50 ohm, has 25 divisions in it. When a current of $4 \times 10^{-4}$ A passes through it, its needle (pointer) deflects by one division. To use this galvanometer as a voltmeter of range 2.5 V, it should be connected to a resistance of :

(1) 250 ohm (2) 200 ohm
(3) 6200 ohm (4) 6250 ohm

**12 Jan Evening**

**Q25**

The charge on a capacitor plate in a circuit, as a function of time, is shown in the figure:




What is the value of current at t = 4 s ?

(1) Zero (2) 3 μA
(3) 2 μA (4) 1.5 μA

**12 Jan Evening**

**Q26**

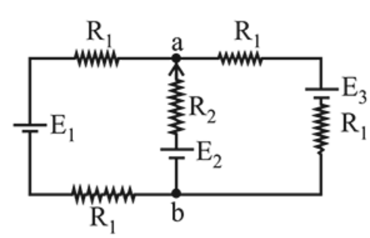
A 200 Ω resistor has a certain color code. If one replaces the red color by green in the code, the new resistance will be :

(1) 100 Ω (2) 400 Ω (3) 300 Ω (4) 500 Ω

**8 April Morning**

**Q27**

For the circuit shown, with $R_1 = 1.0\ \Omega$, $R_2 = 2.0\ \Omega$, $E_1 = 2$ V and $E_2 = E_3 = 4$ V, the potential difference between the points '$a$' and '$b$' is approximately (in V) :

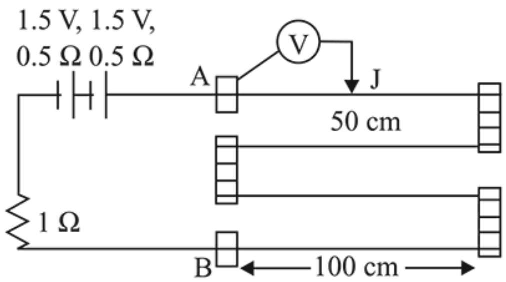



(1) 2.7 (2) 2.3 (3) 3.7 (4) 3.3

**8 April Morning**

**Q28**

In the circuit shown, a four-wire potentiometer is made of a 400 cm long wire, which extends between A and B. The resistance per unit length of the potentiometer wire is $r = 0.01\ \Omega/\text{cm}$. If an ideal voltmeter is connected as shown with jockey J at 50 cm from end A, the expected reading of the voltmeter will be :




(1) 0.50 V (2) 0.75 V (3) 0.25 V (4) 0.20 V

**8 April Evening**

**Q29**

A cell of internal resistance $r$ drives current through an external resistance R. The power delivered by the cell to the external resistance will be maximum when :

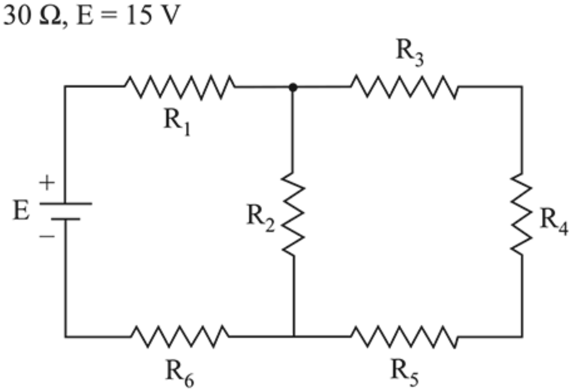
(1) $R = 0.001\ r$ (2) $R = 1000\ r$
(3) $R = 2r$ (4) $R = r$

**8 April Evening**

**Q30**

In the figure shown, what is the current (in Ampere) drawn from the battery? You are given :

$R_1 = 15\ \Omega$, $R_2 = 10\ \Omega$, $R_3 = 20\ \Omega$, $R_4 = 5\ \Omega$, $R_5 = 25\ \Omega$, $R_6 = 30\ \Omega$, $E = 15$ V

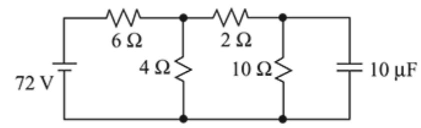



(1) 13/24 (2) 7/18 (3) 9/32 (4) 20/3

**8 April Evening**

**Q31**

Determine the charge on the capacitor in the following circuit:

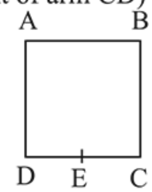



(1) 60 μC (2) 2 μC
(3) 10 μC (4) 200 μC

**9 April Morning**

**Q32**

A wire of resistance R is bent to form a square ABCD as shown in the figure. The effective resistance between E and C is: (E is mid-point of arm CD)




(1) R (2) $\frac{7}{64}R$ (3) $\frac{3}{4}R$ (4) $\frac{1}{16}R$

**9 April Morning**

**Q33**

A moving coil galvanometer has resistance 50 Ω and it indicates full deflection at 4 mA current. A voltmeter is made using this galvanometer and a 5 k Ω resistance. The maximum voltage, that can be measured using this voltmeter, will be close to:

(1) 40 V (2) 15 V (3) 20 V (4) 10 V

**9 April Morning**

**Q34**

A metal wire of resistance 3 Ω is elongated to make a uniform wire of double its previous length. This new wire is now bent and the ends joined to make a circle. If two points on the circle make an angle 60° at the centre, the equivalent resistance between these two points will be:

(1) $\frac{12}{5}\Omega$ (2) $\frac{5}{2}\Omega$ (3) $\frac{5}{3}\Omega$ (4) $\frac{7}{2}\Omega$

**9 April Evening**

**Q35**

The resistance of a galvanometer is 50 ohm and the maximum current which can be passed through it is 0.002 A. What resistance must be connected to it order to convert it into an ammeter of range 0 – 0.5 A?

(1) 0.5 ohm (2) 0.002 ohm
(3) 0.02 ohm (4) 0.2 ohm

**9 April Evening**

### Q36

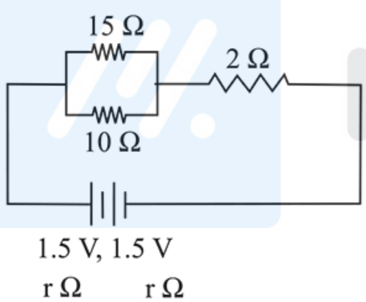
In a conductor, if the number of conduction electrons per unit volume is $8.5 \times 10^{28}\ m^{-3}$ and mean free time is 25 *fs* (femto second), it's approximate resistivity is: ($m_e = 9.1 \times 10^{-31}$ kg)

(1) $10^{-6}\ \Omega m$ (2) $10^{-7}\ \Omega m$ (3) $10^{-8}\ \Omega m$ (4) $10^{-5}\ \Omega m$

**9 April Evening**

### Q37

In the given circuit, an ideal voltmeter connected across the 10 Ω resistance reads 2V. The internal resistance r, of each cell is :

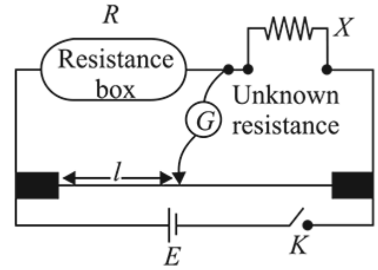



(1) 1 Ω (2) 0.5 Ω (3) 1.5 Ω (4) 0 Ω

**10 April Morning**

### Q38

In a meter bridge experiment, the circuit diagram and the corresponding observation table are shown in figure.




| Sl.No. | R \| | l (cm) |
|---|---|---|
| 1. | 1000 | 60 |
| 2. | 100 | 13 |
| 3. | 10 | 1.5 |
| 4. | 1 | 1.0 |

Which of the reading is consistent ?

(1) 3 (2) 2 (3) 4 (4) 1

**10 April Morning**

### Q39

A current of 5 A passes through a copper conductor (resistivity) = $1.7\times10^{-8}\Omega m$) of radius of cross-section 5 mm. Find the mobility of the charges if their drift velocity is $1.1\times10^{-3}$ m/s.

(1) $1.8 m^2/Vs$ (2) $1.5\ m^2/Vs$
(3) $1.3\ m^2/Vs$ (4) $1.0\ m^2/Vs$

**10 April Morning**

### Q40

Space between two concentric conducting spheres of radii *a* and *b* ($b > a$) is filled with a medium of resistivity *p*. The resistance between the two spheres will be :

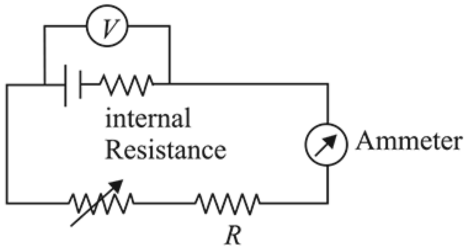
(1) $\frac{p}{4\pi}\left(\frac{1}{a}-\frac{1}{b}\right)$ (2) $\frac{p}{2\pi}\left(\frac{1}{a}-\frac{1}{b}\right)$

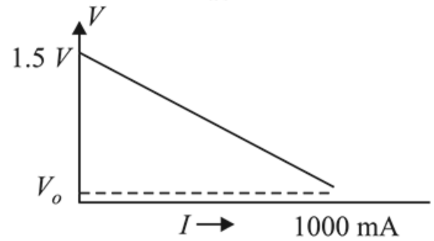
(3) $\frac{p}{2\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$ (4) $\frac{p}{4\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$

**10 April Evening**

### Q41

To verify Ohm's law, a student connects the voltmeter across the battery as, shown in the figure. The measured voltage is plotted as a function of the current, and the following graph is obtained :




If $V_o$ is almost zero, identify the correct statement:

(1) The emf of the battery is 1.5 V and its internal resistance is 1.5 Ω

(2) The value of the resistance R is 1.5 Ω

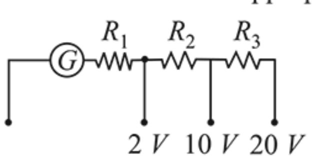
(3) The potential difference across the battery is 1.5 V when it sends a current of 1000 mA

(4) The emf of the battery is 1.5 V and the value of R is 1.5 Ω

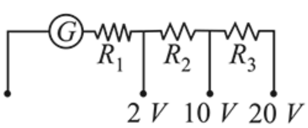
**12 April Morning**

### Q42

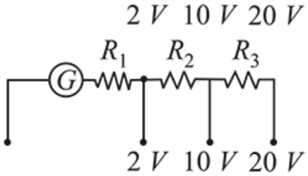
A galvanometer of resistance 100 Ω has 50 divisions on its scale and has sensitivity of 20 μA/division. It is to be converted to a voltmeter with three ranges, of 0–2V, 0–10 V and 0–20 V. The appropriate circuit to do so is :

(1) 

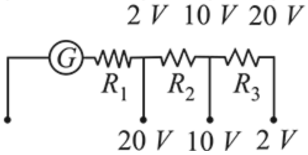
$R_1 = 2000\ \Omega$, $R_2 = 8000\ \Omega$, $R_3 = 10000\ \Omega$

(2) 

$R_1 = 1900\ \Omega$, $R_2 = 9900\ \Omega$, $R_3 = 19900\ \Omega$

(3) 

$R_1 = 1900\ \Omega$, $R_2 = 8000\ \Omega$, $R_3 = 10000\ \Omega$

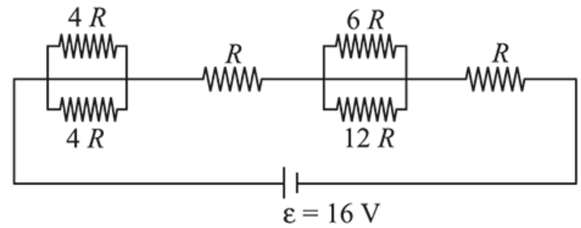
(4) 

$R_1 = 19900\ \Omega$, $R_2 = 9900\ \Omega$, $R_3 = 1900\ \Omega$

**12 April Morning**

### Q43

The resistive network shown below is connected to a D.C. source of 16 V. The power consumed by the network is 4 Watt. The value of R is :




(1) 6Ω (2) 8 Ω (3) 1 Ω (4) 16 Ω

**12 April Morning**

### Q44

A moving coil galvanometer, having a resistance G, produces full scale deflection when a current $I_g$ flows through it. This galvanometer can be converted into (i) an ammeter of range 0 to $I_0$ ($I_0 > I_g$) by connecting a shunt resistance $R_A$ to it and (ii) into a voltmeter of range 0 to V ($V = GI_0$) by connecting a series resistance $R_v$ to it. Then,

(1) $R_A R_V = G^2\left(\frac{I_0 - I_g}{I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

(2) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

(3) $R_A R_V = G^2\left(\frac{I_g}{I_0 - I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g}\right)^2$

(4) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \frac{I_g}{(I_0 - I_g)}$

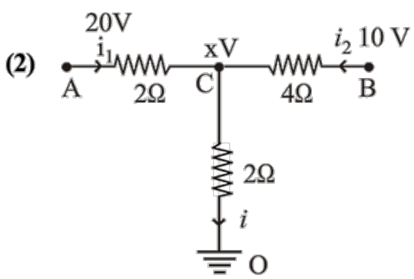
**12 April Evening**

# JEE Mains 2019 Chapter wise Question Bank

## Current Electricity - Answers

**Q1**

(2)




Let voltage at C = xV

From kirchhoff's current law,

KCL : $i_1 + i_2 = i$

$$\frac{20-x}{2}+\frac{10-x}{4}=\frac{x-0}{2} \Rightarrow x=10$$

$$\therefore i=\frac{V}{R}=\frac{X}{R}=\frac{10}{2}=5A$$

**9 Jan Morning**

**Q2**

(2) Color code : Bl, Br, R, O, Y, G, B, V, Gr, W

0, 1, 2, 3, 4, 5, 6, 7, 8 9

$R = AB \times C \pm D\%$ where D = tolerance

$D_{gold} = \pm 5\%, D_{silver} = \pm 10\%; D_{no\ colour} = \pm 20\%$

Red voilet orange silver

$R = 27 \times 10^3\ \Omega \pm 10\%$

$i = 27\ k\Omega \pm 10\%$

**9 Jan Morning**

**Q3**

(1) Using, $I = neAv_d$

$$\therefore \text{Drift speed } v_d = \frac{1}{neA}$$

$$= \frac{1.5}{9\times10^{28}\times1.6\times10^{-19}\times5\times10^{-6}} = 0.02\ mm/s^{-1}$$

**9 Jan Morning**

**Q4**

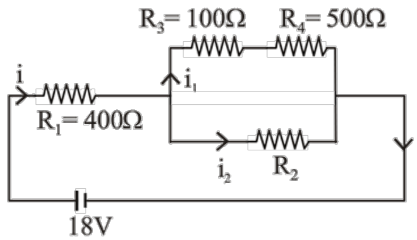
(1) Colour code for carbon resistor
Bl, Br, R, O, Y, G, Blue, V, Gr, W
0 1 2 3 4 5 6 7 8 9
Resistance, $R = AB \times C \pm D$
Bands A and B are the first two significant figures of resistance
B and C indicates the decimal multiplier or the number of zeros that follow A and B
B and D is tolerance: Gold = ± 5%,
Silver = ± 10 % No colour = ± 20%

$R = 53 \times 10^4 \pm 5\% = 530\,k\Omega \pm 5\%$

**9 Jan Evening**

**Q5**

**(1)**




Across $R_4$ reading of voltmeter, $V_4 = 5V$

Current, $i_4 = \frac{V_4}{R_4} = 0.01A$

$V_3 = i_1 R_3 = 1V$
$V_3 + V_4 = 6V = V_2$
$V_1 + V_3 + V_4 = 18V$
$\Rightarrow V_1 = 12\,V$

$i = \frac{V_1}{R_1} = 0.03A$

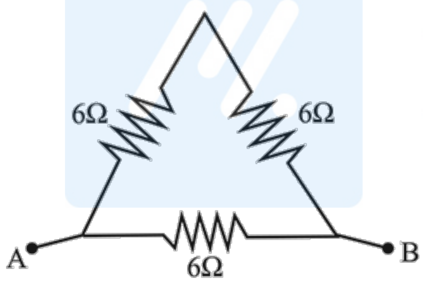
$i = i_1 + i_2 \Rightarrow i_2 = i - i_1 = 0.03 - 0.01A = 0.02\,A$

$\therefore R_2 = \frac{V_2}{i_2} = \frac{6}{0.02} = 300\Omega$

**9 Jan Evening**

**Q6**

**(1)**




Resistance, $R \propto l$ so resistance of each side of the equilateral triangle = 6 Ω

Resistance $R_{eq}$ between any two vertices

$$\frac{1}{R_{eq}} = \frac{1}{12} + \frac{1}{6} \Rightarrow R_{eq.} = 4\Omega$$

**10 Jan Morning**

**Q7**

**(1)** Colour code for carbon resistor

Bl, Br, R, O, Y, G, Blue, V, Gr, W
0 1 2 3 4 5 6 7 8 9

Resistance, $R = AB \times C \pm D$

$\therefore$ Resistance, $R = 50 \times 10^2\,\Omega$

Now using formula, Power, $P = i^2R$

$$\therefore i = \sqrt{\frac{P}{R}} = \sqrt{\frac{2}{50 \times 10^2}} = 20mA$$

**10 Jan Morning**

**Q8**

**(3)** Let x be the length AJ at which galvanometer shows null deflection current,

$$i = \frac{\varepsilon}{12r + r} = \frac{3}{13r}$$

$$\text{or, } i\left(\frac{x}{L}12r\right) = \frac{\varepsilon}{2}$$

$$\Rightarrow \frac{\varepsilon}{13r}\left[\frac{x}{L}\cdot 12r\right] = \frac{\varepsilon}{2}$$

$$\text{or, } x = \frac{13L}{24}$$

**10 Jan Morning**

**Q9**

**(3)** Current passing through resistance $R_1$,

$$i_1 = \frac{v}{R_1} = \frac{10}{20} = 0.5A \text{ and, } i_2 = 0$$

**10 Jan Morning**

**Q10**

**(2)** using, $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

$0.95\,R = \frac{R R\upsilon}{R + R\upsilon}$ (measured value 5% less then internal resistance of voltmeter)

or, $0.95 \times 30 = 0.05\,R\upsilon$

$\therefore\ R\upsilon = 19 \times 30 = 570\,\Omega$

**10 Jan Evening**

**Q11**

**(1)** Power, $P = I^2R$

$4.4 = 4 \times 10^{-6} \times R$

$\Rightarrow R = 1.1 \times 10^6\Omega$

When supply of 11 v is connected

Power, $P' = \frac{v^2}{R} = \frac{11^2}{1.1} \times \frac{11^2}{1.1} \times 10^{-6}$

$= 11 \times 10^{-5}$ W

**10 Jan Evening**

**Q11**

**(1)** Given, colour code of resistance,

$R_1$ = Orange, Red and Brown

$\therefore R_1 = 32 \times 10 = 320$

using balanced wheatstone bridge principle,

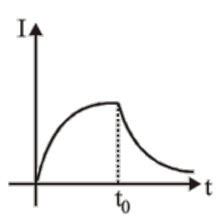
$$\frac{R_1}{R_3} = \frac{R}{R_4} \Rightarrow \frac{320}{R_3} = \frac{80}{40}$$

$\therefore R_3 = 160$ i.e. colour code for $R_3$ Brown, Blue and Brown

**10 Jan Evening**

**Q12**

**(2)**




The current will grow for the time $t = 0$ to $t = t_0$ and after that decay of current takes place.

**11 Jan Morning**

**Q13**

**(2)** Given, Emf of cell, $\varepsilon = 0.5$ v

Rheostat resistance, $R_h = 2\Omega$

Potential gradient is

$$\frac{dv}{dL} = \left(\frac{6}{2+4}\right) \times \frac{4}{L}$$

Let null point be at $\ell$ cm when cell of emf $\varepsilon = 0.5$ v is used.

thus $\varepsilon_1 = 0.5V = \left(\frac{6}{2+4}\right) \times \frac{4}{L} \times \ell$ ... (i)

For resistance $R_h = 6\Omega$ new potential gradient is

$\left(\frac{6}{4+6}\right) \times \frac{4}{L}$ and at null point

$\left(\frac{6}{4+6}\right)\left(\frac{4}{L}\right) \times \ell = \varepsilon_2$ ... (ii)

Dividing equation (i) by (ii) we get

$\frac{0.5}{\varepsilon_2} = \frac{10}{6}$ thus $\varepsilon_2 = 0.3$

**11 Jan Morning**

**Q14**

**(2)** When two resistances are connected in series,

$R_{eq} = 2R$

Power consumed, $P = \frac{\varepsilon^2}{R_{eq}} = \frac{\varepsilon^2}{2R}$

In parallel condition, $R_{eq} = R/2$.

New power, $P' = \frac{\varepsilon^2}{(R/2)}$

or $P' = 4P = 240$ W $(\because P = 60$ W$)$

**11 Jan Morning**

**Q14**

**(4)**

**11 Jan Morning**

**Q15**

**(2)** Given, $E_1 = 1V$, $E_2 = 2V$, $E_3 = 3V$, $r_1 = 1\Omega$, $r_2 = 1\Omega$ and $r_3 = 1\Omega$

$$V_{AB} = V_{CD} = \frac{\frac{E_1}{r_1}+\frac{E_2}{r_2}+\frac{E_3}{r_3}}{\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}} = \frac{\frac{1}{1}+\frac{2}{1}+\frac{3}{1}}{\frac{1}{1}+\frac{1}{1}+\frac{1}{1}}$$

$$= \frac{6}{3} = 2V$$

**11 Jan Evening**

**Q16**

**(3)** Deflection current

$= Ig_{max} = nxk = 0.005 \times 30$

Where, n = Number of divisions = 30 and k = 0.005 amp/division

$= 15 \times 10^{-2} = 0.15$

$v = I_g[20 + R]$

$15 = 0.15\,[20 + R]$

$100 = 20 + R$

$R = 80\ \Omega$

**11 Jan Evening**

**Q17**

**(3)** Initially at null deflection $\frac{R_1}{R_2} = \frac{2}{3}$ ...(i)

Finally at null deflection, when null point is shifted

$$\frac{R_1+10}{R_2} = 1 \Rightarrow R_1 + 10 = R_2 \text{ ...(ii)}$$

Solving equations (i) and (ii) we get

$$\frac{2R_2}{3} + 10 = R_2$$

$$10 = \frac{R_2}{3} \Rightarrow R_2 = 30\Omega$$

& $R_1 = 20\Omega$

Now if required resistance is R then

$$\frac{\frac{30\times R}{30+R}}{30} = \frac{2}{3}$$

$R = 60\Omega$

**11 Jan Evening**

**Q18**

**(1)** As $R = \frac{V^2}{P}$, so $R_1 = \frac{220^2}{25}$ and $R_2 = \frac{220^2}{100}$

$$\text{Current flown } i = \frac{220}{R_1 + R_2}$$

$$P_1 = i^2R_1 = \frac{220^2}{\left(\frac{220^2}{25}+\frac{220^2}{100}\right)} \times \frac{220^2}{25} = 16\text{ W}$$

Similarly, $P_2 = i^2R_2 = 4\text{ W}$

**12 Jan Morning**

**Q19**

**(2)** When key $K_1$ is closed and key $K_2$ is open

$$i_g = \frac{E}{220+R_g} = C\theta_0 \quad \ldots (i)$$

When both the keys are closed

$$i_g = \left(\frac{E}{220+\frac{5R_g}{5+R_g}}\right) \times \frac{5}{(R_g+5)} = \frac{C\theta_0}{5}$$

$$\Rightarrow \frac{5E}{225R_g+1100} = \frac{C\theta_0}{5} \quad \ldots (ii)$$

$$\frac{E}{220+R_g} = C\theta_0 \quad \ldots (i)$$

Dividing (i) by (ii), we get

$$\Rightarrow \frac{225R_g+1100}{1100+5R_g} = 5$$

$$\Rightarrow 5500 + 25R_g = 225R_g + 1100$$

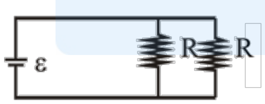
$$200R_g = 4400$$

$$R_g = 22\Omega$$

**12 Jan Morning**

**Q20**

**(4)** Long time after switch is closed, the inductor will be idle so, the equivalent diagram will be as below




$$I = \frac{\varepsilon}{\left(\frac{R\times R}{R+R}\right)} = \frac{2\varepsilon}{R} = \frac{2\times 15}{5} = 6\,A$$

**12 Jan Morning**

**Q21**

**(3)** We have given

$$\frac{dR}{d\ell} \propto \frac{1}{\sqrt{\ell}} \Rightarrow \frac{dR}{d\ell} = k \times \frac{1}{\sqrt{\ell}} \quad \text{(where k is constant)}$$

$$dR = k\frac{d\ell}{\sqrt{\ell}}$$

Let $R_1$ and $R_2$ be the resistance of AP and PB respectively.

Using wheatstone bridge principle

$$\therefore \frac{R'}{R'} = \frac{R_1}{R_2} \text{ or } R_1 = R_2$$

$$\text{Now, } \int dR = k\int \frac{d\ell}{\sqrt{\ell}}$$

$$\therefore R_1 = k\int_0^{\ell} \ell^{-1/2} d\ell = k.2.\sqrt{\ell}$$

$$R_2 = k\int_{\ell}^{1} \ell^{-1/2} d\ell = k.(2-2\sqrt{\ell})$$

Putting $R_1 = R_2$

$$k2\sqrt{\ell} = k(2-2\sqrt{\ell})$$

$$\therefore 2\sqrt{\ell} = 1$$

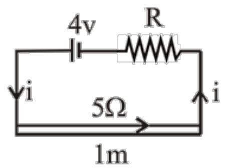
$$\sqrt{\ell} = \frac{1}{2}$$

$$\text{i.e., } \ell = \frac{1}{4}\text{m} \Rightarrow 0.25\text{ m}$$

**12 Jan Morning**

**Q22**

**(3)**




Current flowing through the circuit (I) is given by

$$I = \left(\frac{4}{R+5}\right) A$$

Resistance of length 10 cm of wire

$$= 5 \times \frac{10}{100} = 0.5\Omega$$

According to question,

$$5 \times 10^{-3} = \left(\frac{4}{R+5}\right).(0.5)$$

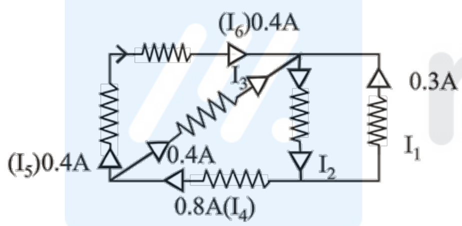
$$\therefore \frac{4}{R+5} = 10^{-2} \text{ or } R+5 = 400\Omega$$

$$\therefore R = 395\Omega$$

**12 Jan Morning**

### Q23

**(2)**




From KCL, $I_3 = 0.8 - 0.4 = 0.4$ A

$I_2 = 0.4 + 0.4 + 0.3$

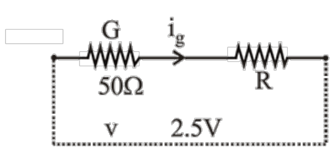
$= 1.1$ A

and $I_6 = 0.4$ A

**12 Jan Evening**

### Q24

(2) Galvanometer has 25 divisions $I_g = 4 \times 10^{-4} \times 25 = 10^{-2}$ A




$V = I_g R_{net}$

$v = I_g (G + R)$

$2.5 = (50 + R)10^{-2} \therefore R = 200\Omega$

**12 Jan Evening**

### Q25

**(1)** Clearly, from graph

Current, $I = \frac{dq}{dt} = 0$ at $t = 4$s [Since q is constant]

**12 Jan Evening**

### Q26

**(4)** Number 2 is associated with the red colour. This colour is replaced by green.

$\because$ Colour code figure for green is 5

$\therefore$ New resistance = 500 Ω

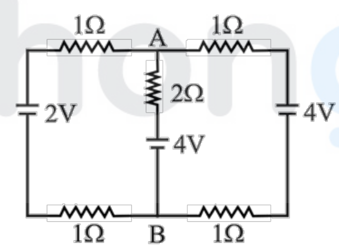
**8 April Morning**

### Q27

**(4)** Applying parallel combination of batteries

Equivalent emf between A and B is

$$= \frac{\frac{E_1}{1+1} + \frac{E_2}{2} + \frac{E_3}{1+1}}{\frac{1}{1+1} + \frac{1}{2} + \frac{1}{1+1}}$$




$$\frac{\frac{2}{2} + \frac{4}{2} + \frac{4}{2}}{\frac{1}{2} + \frac{1}{2} + \frac{1}{2}} = \frac{5 \times 2}{3}$$

$$= \frac{10}{3} = 3.3 \text{ Volt}$$

**8 April Morning**

### Q28

**(3)** The resistance of potentiometer wire

$R = 0.01 \times 400 = 4\ \Omega$

Current in the wire

$$i = \frac{V}{R_T} = \frac{3}{4 + 0.5 + 0.57 + 1} = \frac{1}{2} A$$

Now $V = iR_{AJ} = \frac{1}{2} \times (0.01 \times 50) = 0.25$ V.

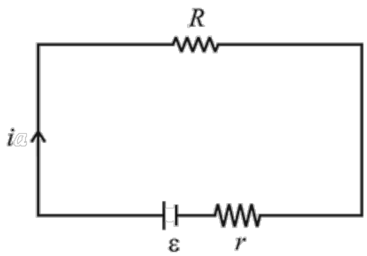
**8 April Evening**

**Q29**

(4) $i = \left(\dfrac{\varepsilon}{R+r}\right)$

Power delivered to R.

$P = i^2 R = \left(\dfrac{\varepsilon}{R+r}\right)^2 R$




P to be maximum, $\dfrac{dP}{dR} = 0$

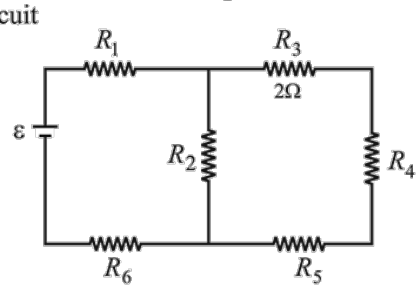
or $\dfrac{d}{dR}\left[\left(\dfrac{\varepsilon}{R+r}\right)^2 R\right] = 0$

or $R = r$

**8 April Evening**

**Q30**

(3) $R_3$, $R_4$ and $R_5$ are in series so their equivalent $R = 20 + 5 + 25 = 50\ \Omega$
This is parallel with $R_2$, and so net resistance of the circuit




$R_{eq} = \left(\dfrac{10 \times 50}{10+50}\right) + 15 + 30 = \dfrac{160}{3}\Omega$

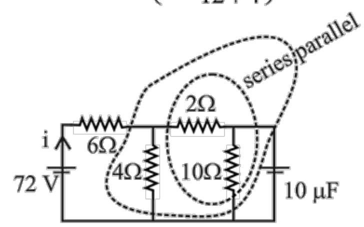
So, $i = \dfrac{\varepsilon}{R_{eq}} = \dfrac{15}{(100/3)} = \dfrac{9}{32}A$

**8 April Evening**

**Q31**

(4) At steady state, there is no current in capacitor.
$2\Omega$ and $10\Omega$ are in series. There equivalent resistance is $12\Omega$. This $12\Omega$ is in parallel with $4\Omega$ and there combined resistance is $12 \times 4/(12+4)$. This resistance is in series with $6\Omega$. Therefore, current drawn from battery

$i = \dfrac{V}{R} = \left(\dfrac{72}{6 + \dfrac{12 \times 4}{12+4}}\right) = 8A$




Current in $10\Omega$ resistor

$i' = \left(\dfrac{4}{4+12}\right)8 = 2A$

Pd across capacitor, $V = i'R = 2 \times 10 = 20V$

$\therefore$ Charge on the capacitor, $q = CV$
$= 10 \times 20 = 200\ \mu C$.

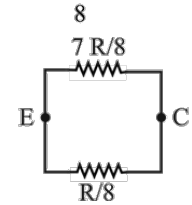
**9 April Morning**

**Q32**

(2) Here $R_{DA} = R_{AB} = R_{BC} = R/4$
and $R_{DE} = R_{EC} = R/8$
Now $R_{ED}, R_{DA}, R_{AB}, R_{BC}$ are in series.

$\therefore R_s = \dfrac{R}{8} + \dfrac{R}{4} + \dfrac{R}{4} + \dfrac{R}{4} = \dfrac{R + 2R + 2R + 2R}{8} = \dfrac{7R}{8}$

$\therefore R_{eq} = \dfrac{\left(\dfrac{7R}{8}\right)\left(\dfrac{R}{8}\right)}{R} = \dfrac{7R}{64}$




**9 April Morning**

**Q33**

(3) $V = i_g(G + R) = 4 \times 10^{-3}$
$(50 + 5000) = 20V$

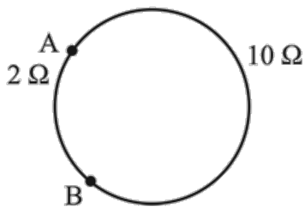
**9 April Morning**

**Q34**

**(3)** When length becomes double its resistance becomes $(R \propto l^2)$

$R = 4 \times 3 = 12\ \Omega$

$$R_{eq} = \frac{2 \times 10}{12} = \frac{5}{3}\Omega$$




**9 April Evening**

**Q35**

**(4)** Using, $i_g = i\frac{S}{S+G}$

$$0.002 = 0.5\frac{S}{S+50}$$

On solving, we get

$$S = \frac{100}{498} \simeq 0.2\Omega$$

**9 April Evening**

**Q36**

**(3)** $\rho = \frac{m}{ne^2\tau}$

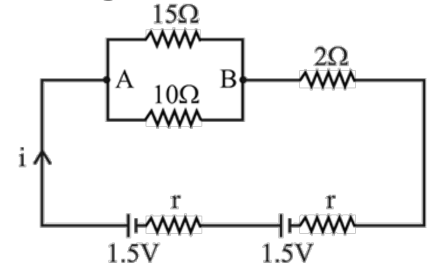
$$= \frac{9.1 \times 10^{-31}}{8.5 \times 10^{28} \times (1.6 \times 10^{-19})^2 \times 25 \times 10^{-15}}$$

$= 10^{-8}\ \Omega$-m

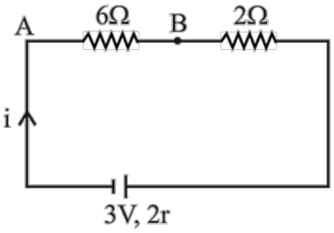
**9 April Evening**

**Q37**

(2) For the given circuit




$\equiv$




$$i = \frac{3}{8+2r}$$

Now voltage across AB

$$i \times 6 = \frac{3}{8+2r} \times 6 = 2$$

$\Rightarrow 9 = 8 + 2r$

$\Rightarrow r = \frac{1}{2}\Omega$

**10 April Morning**

**Q38**

**(3)** For a balanced bridge

$$\frac{R_1}{R_2} = \frac{l_2}{l_1}$$

So $\frac{R}{X} = \frac{l}{100-l}$

Using the above expression

$$X = \frac{R(100-l)}{l}$$

for observation (1) $X = \frac{100 \times 40}{60} = \frac{2000}{3}\Omega$

for observation (2) $X = \frac{100 \times 87}{13} = \frac{8700}{13}\Omega$

for observation (3) $X = \frac{10 \times 98.5}{1.5} = \frac{1970}{3}\Omega$

for observation (4) $X = \frac{1 \times 99}{1} = 99\Omega$

Clearly we can see that the value of x calculated in observation (4) is inconsistent than other.

**10 April Morning**

**Q39**

**(4)** Charge mobility

$$(\mu) = \frac{V_d}{E} \ \left[\text{Where } V_d = \text{drift velocity}\right]$$

and resistivity $(\rho) = \frac{E}{j} = \frac{EA}{I} \Rightarrow E = \frac{I(\rho)}{A}$

$$\Rightarrow \mu = \frac{V_d}{E} = \frac{V_d A}{I_\rho}$$

$$= \frac{1.1\times10^{-3}\times\pi\times\left(5\times10^{-3}\right)^2}{5\times1.7\times10^{-8}}$$

$$\mu = 1.0\frac{m^2}{V_s}$$

**10 April Morning**

**Q40**

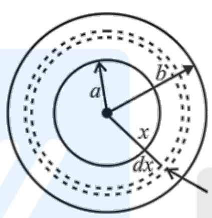
**(1)** $dR = \frac{(\rho)(dx)}{4\pi x^2}$

$R = \int dR$

$\int dR = \rho\int_a^b \frac{dx}{4\pi x^2}$

$\Rightarrow R = \frac{\rho}{4\pi}\left[\frac{-1}{x}\right]_a^b$

$R = \left(\frac{\rho}{4\pi}\right)\cdot\left(\frac{1}{a}-\frac{1}{b}\right)$




**10 April Evening**

**Q41**

**(1)** When $i = 0$, $V = \varepsilon = 1.5$ volt

**12 April Morning**

**Q42**

**(3)** $i_g = 20 \times 50 = 1000\ \mu A = 1$ mA

Using, $V = i_g(G + R)$, we have

$2 = 10^{-3}(100 + R_1)$

$R_1 = 1900\ \Omega$

when, $V = 10$ volt

$10 = 10^{-3}(100 + R_2 + R_1)$

$10000 = (100 + R_2 + 1900)$

$\therefore\ R_2 = 8000\ \Omega$

**12 April Morning**

**Q43**

**(2)** Equivalent resistance,

$$R_{eq} = \frac{4R\times4R}{4R+4R} + R + \frac{6R\times12R}{6R+12R} + R$$
$$= 2R + R + 4R + R$$
$$= 8R.$$

Using, $P = \frac{V^2}{R_{eq}}$

$$4 = \frac{16^2}{8R}$$

$$\therefore\ R = \frac{16^2}{4\times8} = 8\ \Omega$$

**12 April Morning**

**Q44**

**(2)** In an ammeter,

$$i_g = i_0\frac{R_A}{R_A + G}$$

and for voltmeter,

$V = i_g(G + R_V) = Gi_0$

On solving above equations, we get

$R_A R_V = G^2$

and $\frac{R_A}{R_V} = \left(\frac{i_g}{i_0 - i_g}\right)^2$

**12 April Evening**